THE LOCUS OF POINTS WHERE A FINITELY PRESENTED FLAT MODULE IS FLAT OVER AN INTERMEDIATE SCHEME IS OPEN AND COMMUTES WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4701-4720 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4722-4756, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Flatness of a module over a morphism, finite presentation of modules and morphisms, and the open nature of the flat locus; the base-change comparison is retained.

Source reference. Original lemma lemma-flat-locus-base-change: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4468–4486. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-finite-presentation-permanence: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4756–4756. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma theorem-criterion-flatness-fibre: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4590–4599. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma theorem-openness-flatness: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4446–4449. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of schemes over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Assume*

(1) *X is locally of finite presentation over S,*
(2) *$\mathcal{F}$ an $\mathcal{O}_X$-module of finite presentation,*
(3) *$\mathcal{F}$ is flat over S, and*
(4) *Y is locally of finite type over S.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Then the set

$$U = \{x \in X \mid \mathcal{F} \text{ flat at } x \text{ over } Y\}.$$

is open in X and its formation commutes with arbitrary base change: If $S' \to S$ is a morphism of schemes, and U' is the set of points of $X' = X \times_S S'$ where $\mathcal{F}' = \mathcal{F} \times_S S'$ is flat over $Y' = Y \times_S S'$, then $U' = U \times_S S'$.

Proof. By Morphisms, Lemma morphisms-lemma-finite-presentation-permanence the morphism f is locally of finite presentation. Hence U is open by Theorem theorem-openness-flatness. Because we have assumed that $\mathcal{F}$ is flat over S we see that Theorem theorem-criterion-flatness-fibre implies

$$U = \{x \in X \mid \mathcal{F}_s \text{ flat at } x \text{ over } Y_s\}.$$

where s always denotes the image of x in S. (This description also works trivially when $\mathcal{F}_x = 0$.) Moreover, the assumptions of the lemma remain in force for the morphism $f' : X' \to Y'$ and the sheaf $\mathcal{F}'$. Hence U' has a similar description. In other words, it suffices to prove that given $s' \in S'$ mapping to $s \in S$ we have

$$\{x' \in X'_{s'} \mid \mathcal{F}'_{s'} \text{ flat at } x' \text{ over } Y'_{s'}\}$$

is the inverse image of the corresponding locus in X_s. This is true by Lemma lemma-flat-locus-base-change because in the cartesian diagram

$$\begin{array}{ccc} X'_{s'} & \longrightarrow & X_s \\ \downarrow & & \downarrow \\ Y'_{s'} & \longrightarrow & Y_s \end{array}$$

the horizontal morphisms are flat as they are base changes by the flat morphism $\mathrm{Spec}(\kappa(s')) \to \mathrm{Spec}(\kappa(s))$. □